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Question Paper Code : 21559

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2024

Third Semester

Computer Science and Engineering

23CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2023)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

U, CO1 1. What is the the time complexity of the binary search algorithm? Justify your answer.

A) O(n) B) $O(n^2)$

C) O(log n) D) O(2n)

U, CO1 2. What does the following algorithm compute? What is the basic operation?

```
ALGORITHM Secret(A[0..n − 1])
//Input: An array A[0..n − 1] of n real numbers
minval←A[0]; maxval← A[0]
for i← 1 to n − 1 do
  if A[i] <minval
      minval← A[i]
  if A[i] >maxval
     maxval← A[i]
  return maxval–minval
```

A, CO2 3. What is the maximum number of operands in the stack while evaluating the given postfix operation 6 2 3 + - 3 8 2 / + *? Justify your answer.

A) 1 B) 2 C) 3 D) 4

U, CO2 4. In Queue, an element can be inserted and deleted by using _____. Justify your answer.

A) Front and Rear B) Rear and Front

C) Top D) None of the options

U, CO3 5. Implementation of priority queue ADT can be done by using _____data structure. Justify your answer.

A) Binary heap B) Hashing

C) Dequeue D) None of the options

.R, CO3 6. Define the terms Degree of a tree and Height of the tree.

U, CO4 7. Search tables used by compiler for efficient searching generally use _____. Give proper reason.

A) Hash table B) List of records

C) Binary search table D) Binary search tree

R, CO4 8. Write the characteristics of the hash function.

U, CO5 9. Compare Breadth First Search and Depth First Search.

R, CO5 10. List the algorithms used to find Minimum Spanning Tree.

PART B – (4 × 15 = 60 Marks)

A, CO1 11. a) i. For each of the following pairs of functions, indicate whether the first function of each of the following pairs has a lower, same, or higher order of growth than the second function. Justify your answer.

1. n(n+1) and $2000n^2$
2. logn and n! (8 Marks)

A, CO1 11. a) ii. Write the Linear Search algorithm to search a key in a list of numbers and find the Best case, Worst case and Average case Efficiency of the algorithm. (7 Marks)

OR

A, CO1 11. b) i. Consider the following algorithm.

ALGORITHM *Mystery(n)*

//Input: A nonnegative integer *n*

$S \leftarrow 0$

for $i \leftarrow 1$ to n do

$S \leftarrow S + i*i$

return S

1. Compute space complexity.
2. What is its basic operation? Justify your answer.
3. How many times is the basic operation executed?
4. What is the efficiency class of this algorithm? (8 Marks)

A, CO1 11. b) ii. Solve the following recurrence relation:

T(n) = 3T(n-1) if n>0; T(0) = 1 (7 Marks)

U, CO2 12. a) i. Write the pseudocode / routine for Stack ADT operation PUSH, POP, TOP and ISFULL for Array implementation. Illustrate the operations with a neat diagram. (8 Marks)

A, CO2 12. a) ii. Evaluate the following postfix expression using Stack ADT. Draw the diagram to illustrate the evaluation.

6 2 3 + - 38 2 / + * 2 ^ 3 + (7 Marks)

OR

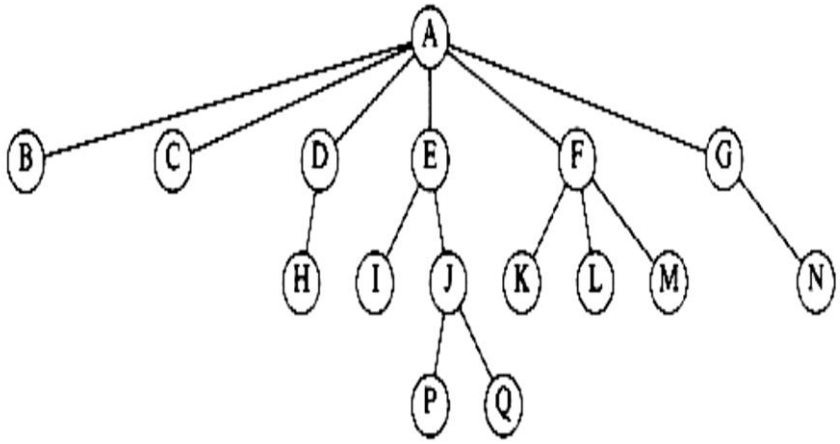
U, CO2 12. b) i. Write the pseudocode / routine for Queue ADT operation Initialize, EnQueue, DeQueue, Front, Rear, IsEmpty, IsFull and Count using capacity, front and rear index for Circular Array implementation. Illustrate the operations with a neat diagram. (8 Marks)

A, CO2 12. b) ii. What is the output of the following snippets of code?

```
Queue q;
Stack s;
s.push(8);
s.push(9);
s.push(10);
q.enqueue(s.pop());
q.enqueue(8);
q.enqueue(9);
s.push(q.dequeue());
s.Top();
s.pop();
s.Top();
s.pop()
s.IsEmpty()
q.Front()
q.Rear()
```

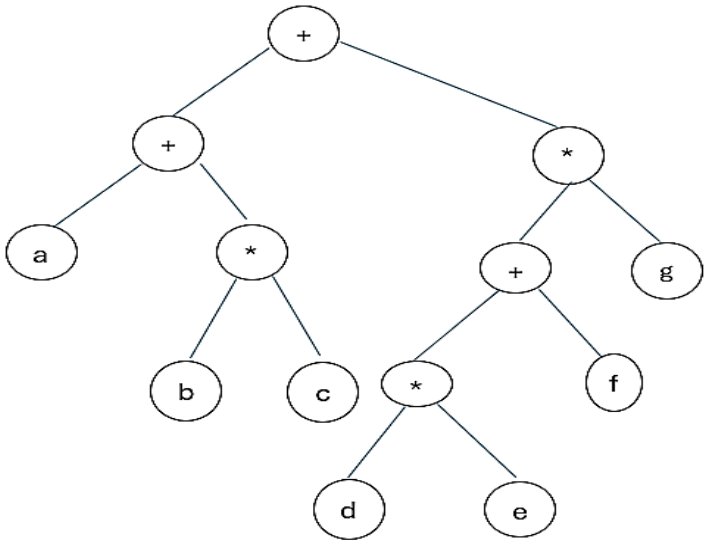
Justify your answer with a diagram illustrating the operation**.** (7 Marks)

U, CO3 13. a) i. Consider the following tree. Identify the root node, leaf nodes, depth of the tree and node has two children after defining the terms. Provide LCRS tree representation by using Linked list for the above tree.



(8 Marks)

A, CO3 13. a) ii. Perform inorder, preorder, postorder and level order traversal of the given tree and output the values.



(7 Marks)

OR

U, CO3 13. b) i. Write and explain insertion algorithm for AVL tree. (8 Marks)

A, CO3 13. b) ii. Construct AVL tree for the nodes given below. Specify the type of rotations involved if any. 2, 4, 1, 42, 10, 12, 9, 22, 15 (7 Marks)

U, CO4 14. a) i. Write short note on Structure property, Heap order property, and the operation percolate up and percolate down related to Binary heap with neat diagram. (8 Marks)

A, CO4 14. a) ii. Construct a binary heap by inserting the following elements one at a time 2, 4, 1, 6, 3, 7, 8, 10, 5, 9. Then perform 3 deletemin operations from the above heap and show the resultant heap. (7 Marks)

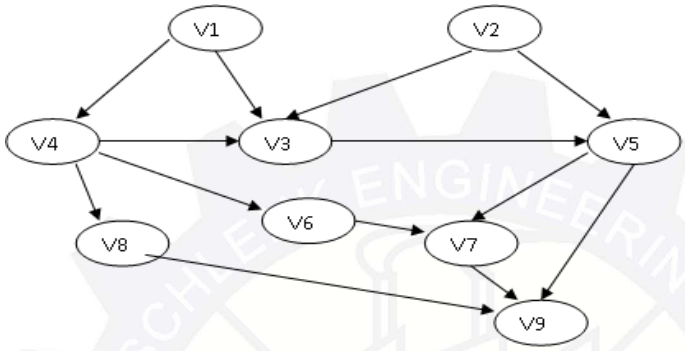
OR

U, CO4 14. b) i. Write short note on double hashing and rehashing. (8 Marks)

A, CO4 14. b) ii. Show the result of inserting the keys 10111101, 00000010, 10011011, 10111110, 01111111, 01010001, 10010110, 00001011, 11001111, 10011110, 11011011, 00101011, 01100001, 11110000, 01101111 into an empty extendible hashing data structure with M=4. (7 Marks)

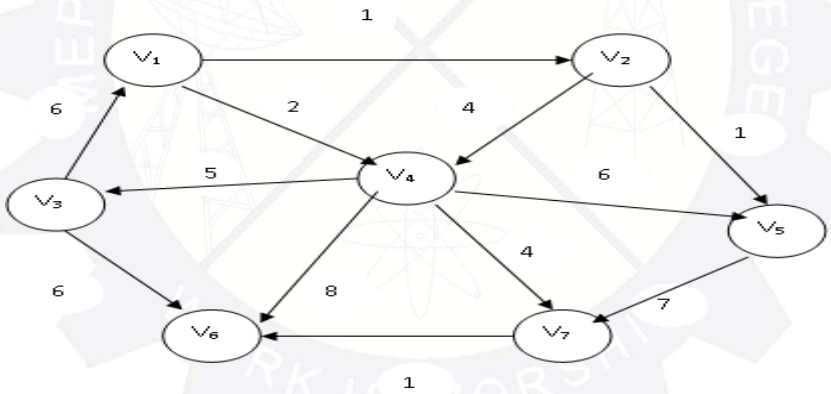
PART C – (1 × 20 = 20 Marks)

A, CO5 15. a) i. Perform the topological sort of the following graph and also provide its algorithm.



(12 Marks)

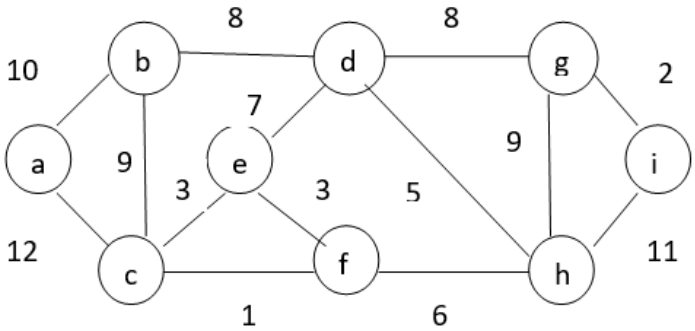
A, CO5 15. a) ii. Apply Dijkstra's algorithm for the following graph:



(8 Marks)

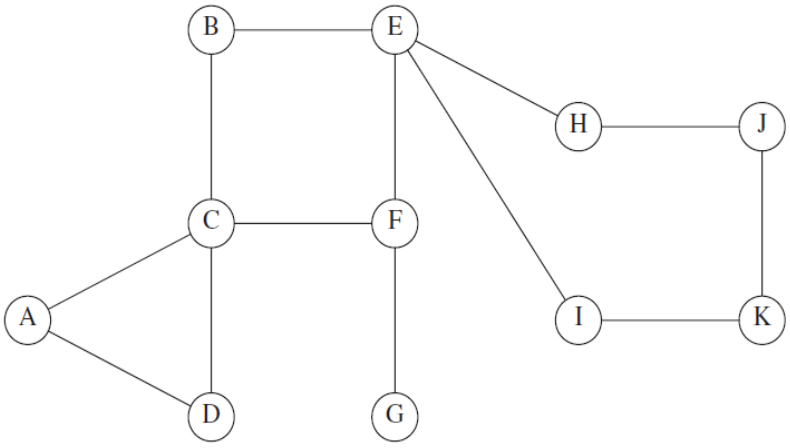
OR

A, CO5 15. b) i. Find Minimum spanning tree using Kruskal's algorithm. Write algorithm for the same.



(12 Marks)

A, CO5 15. b) ii. Find the articulation points in the given graph and also provide its algorithm.



(8 Marks)